# Assessing the effects of wood addition on stream habitat and juvenile coho salmon counts via a BACI Study in Little River, California

by
Emma Legault

A THESIS

submitted to

Oregon State University

Honors College

in partial fulfillment of
the requirements for the
degree of

Honors Baccalaureate of Science in Ecological Engineering
(Honors Scholar)

Presented May 31, 2024
Commencement June 2025

# AN ABSTRACT OF THE THESIS OF

Emma Legault for the degree of Honors Baccalaureate of Science in Ecological Engineering presented on May 31, 2024. Title: Assessing the effects of wood addition on stream habitat and juvenile coho salmon counts via a BACI Study in Little River, California.

Abstract approved:_____

Dana Warren

Throughout the 20th century, coho salmon (Oncorhynchus kisutch) populations and other anadromous salmon in the western US experienced a significant decline. In response to this trend, key sub-populations of coho salmon were listed as threatened under the US Endangered Species Act (ESA) in 1997 and the California Endangered Species Act (CESA) in 2005. Notably, coho salmon decline is intricately linked to habitat loss, particularly the depletion of large wood (LW) the Southern Oregon/Northern California Evolutionary Significant Unit (SONCC ESU). Accordingly, the restoration of LW has become a prevalent method to enhance stream habitat in California and more broadly across streams in the western US. Although large wood restoration activities are prevalent, many projects lack long-term monitoring to gauge their success, and when monitoring is conducted, the complex life history of salmon can complicate the interpretation of results. This study addressed these gaps by presenting assessments on stream habitat and juvenile coho salmon abundance response to LW addition one, three, and five years after treatment using a before-after control-impact (BACI) study design. The BACI study design includes two reaches relatively near each other, where the downstream reach had applied restoration treatment through LW addition, and the upstream reach was unmanipulated. Statistical analyses revealed that there was no significant change in thalweg depth or depth variation. Increased pool area likely contributed to an increase in the density of juvenile coho salmon in the treatment reach in late summer during the first three years after restoration treatment of LW addition. However, during the final fifth year of post-monitoring, low abundances of juvenile coho salmon in both reaches suggest that the restoration work was meeting some aspect of core restoration goals but not all responses persisted. This low abundance year is likely attributed to the substantially earlier surveying period during year 5.

Keywords: BACI study design, Large Wood addition, Habitat enhancement project, coho salmon

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I understand that my project will become part of the permanent collection of Oregon State University, Honors College. My signature below authorizes release of my project to any reader upon request.

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Emma Legault, Author

# Introduction

Throughout the 20th century, coho salmon (*Oncorhynchus kisutch*) populations and other anadromous salmon in the western US experienced a significant decline (Brown et al. 1994, National Marine Fisheries Service and National Marine Fisheries Service, West Coast Region 2014). In response to this trend, key sub-populations of coho salmon were listed as threatened under the US Endangered Species Act (ESA) in 1997 and the California Endangered Species Act (CESA) in 2005. Notably, the Southern Oregon/Northern California Evolutionary Significant Unit (SONCC ESU) emerged as one of the most impacted ESUs, with its decline intricately linked to habitat loss, particularly the depletion of large wood (LW) (Brown et al. 1994). Accordingly, the restoration of LW has become a prevalent method to enhance stream habitat in California and more broadly across streams in the western United States. Although large wood restoration activities are prevalent, many projects lack thorough follow-up assessments to gauge their success, and when assessments are conducted, the complex life history of salmon can complicate the interpretation of results. This study addresses these gaps by presenting assessments on stream habitat and juvenile coho salmon abundance response to LW addition one, three, and five years after treatment using a before-after control-impact (BACI) study design.

The SONCC ESU encompasses coho salmon populations from Cape Blanco, Oregon, to Punta Gorda, California. Despite this extensive range which provides a portfolio of independent populations across the ESU, the National Marine Fisheries Service (NMFS) report in 2014 indicates that over 75% of these populations are at considerable risk of extinction (Brown et al. 1994, National Marine Fisheries Service and National Marine Fisheries Service, West Coast Region 2014). The declines in the populations of Coho and other salmonid species across the Pacific Northwest have been attributed to a range of threats such as timber harvest, logging, stream cleaning, dams, stream channelization, and climatic changes (National Marine Fisheries Service and National Marine Fisheries Service, West Coast Region 2014). Among coastal salmon ESU's, the most common key limiting threat is channelization which results in the simplification and straightening of the stream channel and reduction of off-channel habitat (National Marine Fisheries Service, West Coast Region, 2014).

Coho salmon have an anadromous life cycle. Adults spawn in freshwater streams and juvenile fish rear in these systems for one to two years before they migrate to the ocean and live for an additional one to two years as they mature. Adults then return to their natal streams to spawn. In the freshwater stage of their life cycle, salmon require adequate spawning habitat and slower-moving velocities in deeper pools and side channels to provide cover and minimize energy expenditure (Nickelson 1986, Cole 2001 p. 200). Furthermore, large wood additions add spatial complexity within the channel and supports the natural wood recruitment (Wohl and Goode 2008).

Winter survival for coho salmon has been closely linked with pool and off channel habitat (Nickelson 1986, Cole 2001), and in summer juvenile coho survival are often strongly associated with pool habitat – particularly pool habitat associated with large wood and wood jams that provide deep water and overhead cover (Brown et al. 1994, Roni 2003). By increasing salmon survival and abundance, wood jams and associated pool habitats are also important in

maintaining positive feedback in nutrient supplements from adult salmon. Following spawning, adult salmon carcasses provide essential energy and nutrients to the river ecosystem, benefiting newly hatched salmon by increasing basal resource availability (Cederholm et al. 1999, Schindler et al. 2003). The presence of wood jams and deep pools can help to retain the salmon carcasses and allow valuable marine-derived nutrients to remain in the system nearer to where they spawned even during high-flow events (Strobel et al. 2009). Therefore, the restoration of freshwater habitats, especially through the additions of complex wood structures, is a critical tool to help the survival of Coho and other salmonid populations.

Large wood and wood jams (accumulations of large and small wood) are important structural elements in streams. These wood structures create pools and complex habitat that provide juvenile salmon with cover from predators and visual isolation from conspecifics. These structures also divert water to side channels and flood plains during high flow events thereby dissipating stream energy and creating critical off-channel habitat for rearing juvenile salmon (Penn State Extension, 2023). Before the recognition that wood was important for stream habitat, "stream cleaning" practices involved removing all wood from streams during logging. This was a widespread practice across the western US and has been identified as a major factor linking logging to fish declines (Mellina and Hinch 2009). Beginning in the 1980's, there was increased recognition of the importance of large wood as a structural element for maintaining and creating fish habitat (Pess et al. 2023), and since that time, wood addition has gained prominence in habitat restoration. Because wood loads in most coastal streams of the Pacific Northwest are below their historic levels, there is a general assumption that large wood additions are effective in restoration. However, given the cost associated with stream restoration efforts, this assumption should be tested.

Assessment of stream restoration efforts is widely acknowledged as important but is often underfunded. Most assessments that are done tend to be limited in scope, both temporally and spatially. Studies that have evaluated wood restoration effectiveness in the west have, in some cases, demonstrated clear success, but others have had mixed results (Nagayama and Nakamura 2010). Larger structures such as full-spanning log jams have been found to lead to initial increases in winter rearing habitat for coho salmon and cutthroat trout, which was then linked to significant increases in smolt populations (Solazzi et al. 2000). However, not all responses are identified as significant (Roni et al. 2015), but the importance of publishing these results is essential in strengthening understanding of stream restoration.

Shorter term monitoring may be confounded by natural variations in fish stocks and inadequate time for fish to adapt to their altered environment (Binns and Remmick 1994, Kondolf and Micheli 1995). Longer-term investigations are more limited but have indicated that in some systems habitat improvement devices can maintain deep pools for extended periods, supporting fish survival even during low flow conditions, resulting in sustained increases in coho salmon spawner populations over multiple years (Crispin et al. 1993, Bilby et al. 2024). Long-term assessments with data measuring responses over multiple years post-treatment are rare though. And so too are restoration assessments that use reference reaches to account for natural variation that may otherwise confound or lead to misinterpretation of results in assessments conducted only at the treatment reach.

This study aims to provide a rigorous analysis of coho salmon and habitat response to LW and wood jam additions in a coastal stream in Northern California through a BACI study design. Based on earlier restoration work in this region, we expected that the addition of a series of wood structures along a 310 m (1018 ft) reach of stream would increase maximum stream depths (due to greater scour around wood), increase stream pool habitat (both scour pools and dammed pools) and ultimately, increase abundances of juvenile coho salmon (associated with greater depth and increased pool area). We evaluated these expectations through one year of pre-monitoring and monitoring one, three, and five years of post-treatment wood addition. Furthermore, we quantified changes in these metrics in the restoration (treatment) reach relative to those in an unmanipulated upstream reference reach.

# Methods

## Study Reach and Study Design

This study was conducted on a portion of Little River, a coastal stream in northern California. Little River empties directly into the Pacific Ocean in Humboldt County, California. The mouth of Little River is at coordinates 41.0278 degrees north latitude and 124.1117 degrees west longitude. The mainstem extends approximately 96.7 km (60.1 mi) and drains a watershed area of approximately 118 $km^2$ (45.4 $mi^2$), with elevations ranging from sea level at the mouth to 518 m (1700 ft) in the headwater regions. As of 2005, fish were observed by the California Department of Fish and Game up to 21,711 m (71,232 ft) of the 23,429 m (76,869 ft) survey, with numerous log jams, steep waterfalls, and only one possible anthropogenic barrier along the mainstem (Pacific States Marine Fisheries Commission and California Department of Fish and Wildlife 2019).

This study uses a before-after control-impact (BACI) study design with both treatment and reference reaches relatively close to each other on the same river system. The treatment reach was downstream, and the reference reach was upstream. A BACI study design is crucial in assessing the success of restoration efforts when external factors that influence the study's measured metrics can vary over time. For coastal coho salmon, these factors include variation in ocean conditions (that affect anadromous fish survival), and severe weather events (floods and drought). Surveys were conducted one year before treatment (Year 0) and at three periods of post-monitoring (Year A1, Year A3, Year A5).

The area where the restoration was conducted (Figure 1) was along the mainstem where mean stream bankful widths was 15.5 m (50.9 ft). The treatment reach of Little River was 310 m (1018 ft) long. The unmanipulated reference reach began approximately 75 m (250 ft) upstream of the treatment reach and was designed to be the same length. Stream substrates in both reaches were dominated by cobble and gravel. Redwood forest dominates the watershed (California Department of Fish and Game 2005). The restoration area and overall majority of the watershed is private timberland with some private residences and agriculture near the estuary.

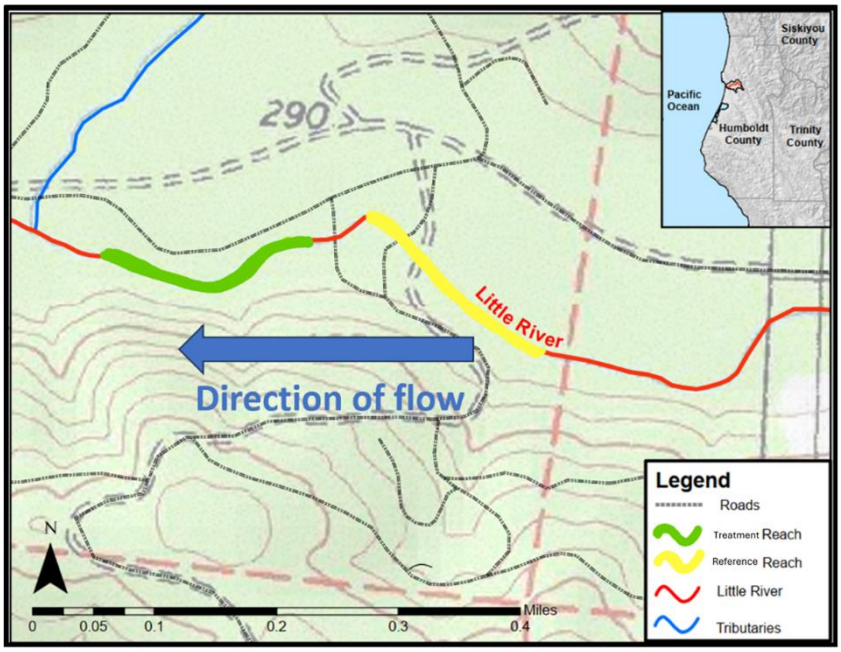


Figure 1. Study area for the Little River BACI study, 2015-2020. Upstream reference reach and downstream treatment reach are both on Little River in Humboldt County in Northern California.

The Little River historically supported strong populations of both coho salmon and chinook salmon, but, as with most streams across the SONCC ESU, returns have declined substantially over the past century. Impacts of logging were documented as early as the mid-1960's, and the system experienced an earthen dam failure in the late 1960's that was estimated to have killed over 20,000 juvenile salmon. Despite this, the highest recorded estimate of Coho smolts (17,843) in the Little River watershed by the Green Diamond Resource Company was in 2022, since trap monitoring began in 2015 (Zontos 2022). This shows promising results contrary to the historic declines since the 1990s but highlights the desire to further protect this threatened species.

The location of the restoration treatment reach on Little River was randomly selected within a larger area of the system that had anadromous fish access. This larger reach was broken into eleven equally spaced transects, perpendicular to the stream, labeled A through K. Transect A represents the most downstream location, while K is the most upstream transect within the reach. The total length of the treatment reach was set at about 20 times the mean bankfull width – based on bankfull width measured at the central stream section F (X site). The distance of each reach transect was set as twice this mean bankfull width. The boundaries of both reaches were adjusted as necessary to begin and end with the start at the edge of a habitat unit (pool or riffle) so as not to end a reach within a unit (Figure 2).

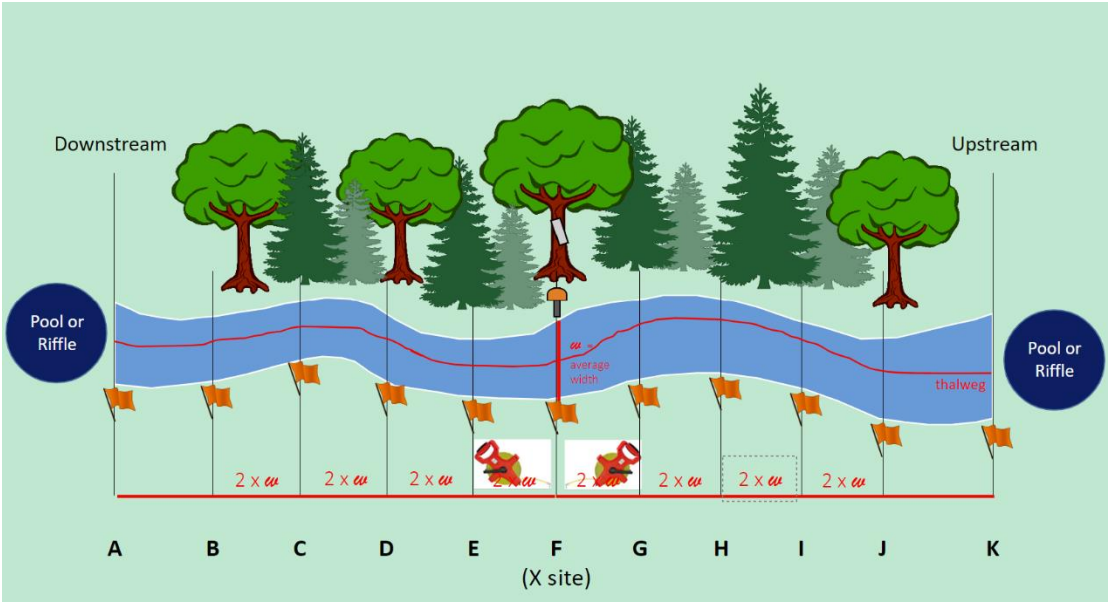


Figure 2. Diagram of protocol for setting up each reach and the transects within them. Obtained from California Department of Fish and Wildlife in their BACI Monitoring Protocol (Pacific States Marine Fisheries Commission and California Department of Fish and Wildlife 2019).

## Restoration Implementation

Restoration in Little River was comprised primarily of large wood additions. In the treatment reach, approximately 13.1 $m^3$ (462.6 $ft^3$) of large wood was added throughout. Along the streambank loom large yellow machines, with their ability to cause destruction but in this case, just the opposite. Piece by piece, wood was carefully placed along the treatment reach only, building structures that would provide habitat to the local fish species migrating back from their tumultuous time in the ocean. The breadth of impact this project has highlights the inspiring nature of large wood restoration.

## Stream habitat and fish assessments

### *Large wood*

Large wood (LW) was defined as dead wood in the bankfull stream channel (or stem at least extends below bankfull) greater than 1.52 m (5 ft) in length and 0.10 m (4 in) in diameter (diameter measured 1/3 up from the base). All LW was enumerated for each reach transect (A-K). LW length and width were measured for an initial set of 10 pieces and then every 5$^{th}$ piece for transects containing 50 to 100 pieces of LW or every 10$^{th}$ piece for transects containing more than 100 pieces of LW thereafter. For the pieces that were not measured directly, lengths and diameters were estimated with the same technician making these estimates for both reaches. For LW that extended across transects, the wood was included only in the transect that had the largest amount of wood in it (to prevent double counting of those pieces).

Each piece was categorized by size class based on the diameter of the large end. Non-cylindrical pieces were approximated for their circular cross-section diameter equivalent in volume. However, there were modifications to length size classes prior to the 2016 (Year A1) surveys. Greater diversity in length size classes added more categories and adjusted the small length class

from less than 3.05 m (10 ft) to less than 5.18 m (17 feet). Length measurements considered only portions of the LW piece with a diameter exceeding 0.10 m (4 inches). Root wads, if present, were not included in measurement estimates.

*Stream depth*

Stream depth measurements were conducted at the same time as the large wood surveys. Within each reach transect, there were 10 evenly spaced cross-sections. At each of the cross sections, the maximum stream depth (thalweg) was recorded along with the habitat unit type (e.g., pool, riffle, cascade, etc.; Table 1)

Table 1. Thalweg Channel and Pool Codes from BACI Monitoring Protocol (Pacific States Marine Fisheries Commission and California Department of Fish and Wildlife 2019).

Table 4. Thalweg Channel and Pool Codes

| POOL FORMING CODES | | CHANNEL UNIT CODES | |
|---|---|---|---|
| N | Not a pool | PP | Pool, Plunge |
| W | Large Woody Debris | PT | Pool, Trench |
| R | Rootwad | PL | Pool, Lateral Scour |
| B | Boulder or Bedrock | PB | Pool, Backwater |
| F | Unknown, Fluvial | PD | Pool, Impoundment |
| | | FW | Flatwater |
| | Combinations eg. WR, BR, WRB | RI | Riffle |
| | | RA | Rapid |
| | | CA | Cascade |
| | | FA | Falls |
| | | DR | Dry Channel |

Source: Peck et. al (2003)

*Pool habitat*

Pool and overall habitat assessments followed California Department of Fish and Wildlife salmonid habitat monitoring protocols (Pacific States Marine Fisheries Commission and California Department of Fish and Wildlife 2019). Briefly, pools were defined as depressions in the streambed that are concave in profile, laterally and longitudinally (see California Department of Fish and Wildlife salmonid habitat monitoring protocols for survey details; (Pacific States Marine Fisheries Commission and California Department of Fish and Wildlife 2019)). The survey focuses on main channel pools where the thalweg runs through the pool (backwater pools were not considered). These pools spanned at least 50% of the wetted channel width at any location. Pools in side channels were not measured. The habitat unit designation at each cross section was used to estimate the total length of that unit in each reach. Adjacent cross sections with the same pool designation were assumed to reflect the same pool and so lengths of both cross sections were added. When a non-pool unit was identified, this represented the end of the pool. If a pool unit was identified at a cross section without any adjacent cross sections identifiers as pools the estimated length was set to the length of one transect (assuming ½ a unit length upstream and downstream from designation). The pool's width was estimated from all the cross-section width measurements within that pool unit. If there were multiple cross sections encompassing a single pool, the mean wetted width from them was used. If a single cross section had a pool, the wetted width from within that cross section was used to estimate the pool width. The estimated length and width of each pool were multiplied together to estimate pool area, and reach-scale pool area was estimated by summing the area of all pools within a reach.

*Fish surveys and reach-scale fish count estimates*

Summer snorkel surveys were conducted to estimate the density of juvenile salmonids and to assess the change in their abundance within each reach and between reaches (Pacific States Marine Fisheries Commission and California Department of Fish and Wildlife 2019). Snorkel surveys are a widely used non-lethal assessment tool to quantify coho salmon in streams. Although in some systems mark-recapture or multiple pass methods depletion with an electronic fisher have been shown to provide slightly more accurate estimates. Given the threatened status of these fish, electrofishing was not deemed appropriate. Given the habitat preferences of juvenile coho salmon for pools, it is assumed here that the snorkel surveys provide an accurate estimation of coho salmon as densities. Further, with a BACI study design, analyses focus on comparisons between paired reaches where the same methods are applied to both areas. Therefore, errors in assessments, if present, are internally consistent and not expected to impact the BACI analysis. Although these surveys were all conducted in summer, there was considerable variation within the season. In most years, surveys were conducted in late summer, but in year 5, at the end of July, three to six weeks earlier than any of the other surveys due to logistical constraints associated with active logging occurring in Little River and California Department of Fish and Wildlife's management of numerous projects during the summer (Table 2). Following restoration treatment, an additional pool formed that was not surveyed.

Table 2. Snorkel survey summary information and percentages of pool area relative to reach area between the treatment and reference reach in Little River.

| Year of study | Surveying date (treatment/reference) | Pools observed (treatment) | Pools surveyed (treatment) | Pools surveyed (reference) | Pool % (treatment) | Pool % (reference) |
|---|---|---|---|---|---|---|
| Year 0 | 8/18/2015 | 3 | 3 | 3 | 61 | 53 |
| Year A1 | 8/31/2016 | 4 | 3 | 3 | 54 | 36 |
| Year A3 | 9/19/2018 | 4 | 3 | 3 | 51 | 34 |
| Year A5 | 7/27/2020 | 4 | 3 | 3 | 58 | 47 |

Snorkel surveys included two people, and the total count for the pool was based on the summed number of fish counted by each individual. Snorkel surveys took place before any other data collection occurred in both reaches. During snorkel surveys, divers identified age 0+ salmonids as: YOY (young of year) coho salmon, YOY chinook Salmon, YOY cutthroat trout, or Unknown YOY salmonids. Unknown salmonids could have included counts of YOY coho salmon but this was unclear likely due to surveying conditions. Thus, without a clear way to know, these values were excluded and represents a potential underestimate of the YOY coho salmon both reaches. Age 1+ individuals (one year or older) were identified as coho salmon, chinook salmon, or cutthroat trout. Given the focus of this restoration on creating juvenile coho salmon habitat, and given the high densities and snorkel count survey focus on age 0+ fish, we focused our analysis on YOY coho salmon, but we also evaluated the total YOY salmonid responses. Age 1+ counts

were inconsistent relative to YOY counts and were deemed not representative of the true counts of the area, thus excluded from this analysis.

During the snorkel surveys, the length of pool surveyed and pool widths at three locations in each pool were recorded. These were used to estimate the area surveyed. Treatment and reference reaches were surveyed on the same day. The snorkel surveys included three to four pools each year, however, based on notes from the field validation data sheets, in a few cases only a portion of the one of the pools was surveyed and in other cases, not all pools in the overall reach were surveyed. Therefore, the total count numbers alone from the snorkel surveys did not reflect the total YOY in each reach (treatment or reference). To estimate total YOY fish from pools in each reach, we used pool area and mean fish densities within pools of each reach. For each reach on each sampling date, we first determined the mean number of YOY coho salmon and YOY salmonids per meter squared of pool habitat surveyed in each pool. We then calculated the average of this YOY fish density in pools for all pools surveys on that date and in each reach (treatment or reference). The mean fish density was then multiplied by the total estimated pool area for the reach during the stream habitat surveys. We used this value as our estimate of overall YOY fish abundance in each reach. Because the snorkel surveys did not necessarily include all pool habitats in a reach during each survey period, total counts alone underestimate total YOY fish densities at the reach scale. The total estimate from mean density and total pool area therefore provided a more accurate whole-reach assessment.

## Data Analysis

*Large Wood (LW) and Thalweg depth*

For stream large wood and stream thalweg depths, we had replicated measurements from each reach transect. Large wood shorter than 3.1 m (10 ft) in length and/or 0.3 m (1 ft) in diameter were identified to cause insignificant impact on pool habitat forming and were excluded from large wood volume calculations. To determine if the restoration influenced total wood volume or thalweg depth relative to the restoration reach over time, we created a linear model that included year, treatment, and a year x treatment interaction:

$$Y \sim \text{Year} + \text{Treatment} + \text{Year:Treatment} + \varepsilon$$

where Y is large wood volume or thalweg depth, Year is a factor for the intervals of data collection before and after treatment (0, 1, 3, 5), treatment is the reach (reference or treatment), Year:Treatment is the interaction term evaluating the BACI effect, and the $\varepsilon$ is the random error We focused on significance of the interaction effect between treatment and year to assess the reference-corrected responses in wood loading or thalweg depth following restoration. Standard deviation of mean thalweg depth for each reach and year was determined through averaging the standard deviation of mean thalweg depth for each transect within the reach.

*Pool area and fish counts*

For pool area estimates, we had whole-reach totals with no data on variation around these estimates. Therefore, parametric statistics could not be conducted, and analysis of those responses were restricted to inferences based on trends over time. We evaluated the totals and, as with LW and thalweg depth, the difference of response in the treatment to reference reach over

time to evaluate relative changes attributable to the wood addition. Increases in these differences relative to the pre-treatment value were attributed to the restoration.

For fish, we had a total estimate for the reach and evaluated relative change due to the restoration using differences between the treatment and reference reach over time. For the initial estimates of fish abundance, we used error around the mean fish density per pool to generate estimates of error around the total reach-scale fish counts in pools but did not propagate error in the pre-post comparisons.

# Results

## Large Wood Volume

Total stream large wood volumes in the treatment ranged from a low of 11.51 $m^3$ (406.64 $ft^3$) before restoration to a high of 26.45 $m^3$ (933.91 $ft^3$) after restoration (Year A3). In the reference reach, wood loads also varied but to a slightly lesser degree (from a low of 9.31 $m^3$ (328.98 $ft^3$) in Year A5 to a high of 18.55 $m^3$ (655.07 $ft^3$) in Year A1).

Mean stream large wood volumes in the treatment ranged from a low of 1.15 $m^3$ (40.66 $ft^3$) before restoration to a high of 2.64 $m^3$ (93.39 $ft^3$) after restoration (Year A3). In the reference reach, wood loads also varied but to a slightly lesser degree (from a low of 0.93 $m^3$ (32.89 $ft^3$) in Year A5 to a high of 1.85 $m^3$ (65.51 $ft^3$) in Year A1).

Table 3. BACI Statistical results for Thalweg Depth and LW Volume including estimates, standard error, t-values, p-values, and lower and upper 95% confidence intervals for the mean reach response to year, reach, and large wood treatment.

| | **Model coefficients** | | **Effect sizes** | | **Confidence Interval (CI)** | |
|---|---|---|---|---|---|---|
| | **Estimate** | **Standard Error** | **t-value** | **p-value** | **Lower CI 95%** | **Upper CI 95%** |
| **Thalweg Depth** | | | | | | |
| Intercept | 1.196 | 0.2348 | 5.095 | 2.71E-06 | 0.736 | 1.656 |
| Year (A1) | -0.134 | 0.332 | -0.404 | 0.688 | -0.785 | 0.517 |
| Year (A3) | -0.086 | 0.332 | -0.259 | 0.796 | -0.737 | 0.565 |
| Year (A5) | 0.086 | 0.332 | -0.259 | 0.796 | -0.565 | 0.737 |
| Reach | 0.56 | 0.332 | 1.524 | 0.132 | -0.091 | 1.211 |
| BACI (A1) | -0.127 | 0.4695 | -0.27 | 0.788 | -1.047 | 0.793 |
| BACI (A3) | -0.052 | 0.4695 | -0.111 | 0.912 | -0.972 | 0.868 |
| BACI (A5) | -0.031 | 0.4695 | -0.066 | 0.948 | -0.951 | 0.889 |
| **LW Volume** | | | | | | |
| Intercept | 49.748 | 19.13 | 2.6 | 0.0113 | 12.253 | 87.243 |
| Year (A1) | 15.759 | 27.054 | 0.582 | 0.5621 | -37.267 | 68.785 |
| Year (A3) | -12.736 | 27.054 | -0.471 | 0.6392 | -65.762 | 40.290 |
| Year (A5) | -16.853 | 27.054 | -0.623 | 0.5353 | -69.879 | 36.173 |

| | | | | | | |
|---|---|---|---|---|---|---|
| Reach | -9.084 | 27.054 | -0.336 | 0.738 | -62.110 | 43.942 |
| BACI (A1) | 30.626 | 38.26 | 0.8 | 0.4261 | -44.364 | 105.616 |
| BACI (A3) | 65.463 | 38.26 | 1.711 | 0.0914 | -9.527 | 140.453 |
| BACI (A5) | 49.725 | 38.26 | 1.3 | 0.1979 | -25.265 | 124.715 |

The greatest mean large wood volume in the reference reach occurred in Year A1. Mean large wood volume in the treatment reach was greatest in Year A3, which we attributed to retention of wood in the dams created in the restoration that was likely transported from the upstream reference reach. Overall, in the year before restoration, both reaches had a comparable number of large wood pieces (Figure 2). The first year after restoration, wood abundances increased in both reaches with a larger increase in the treatment reach. However, due to variation in the distribution of wood throughout the reaches (with some transects having higher wood loads and some having almost no wood) the statistical analysis indicated that the changes were not "significant" in Year A1 (p of interaction term of BACI A1 = 0.8; Table 3). That said, the mean large wood volume differences between the reaches clearly shifted in response to the restoration with 0.26 $m^3$ (9.1 $ft^3$) less wood in the treatment reach before to 0.61 $m^3$ (21.5 $ft^3$) more in the treatment reach in year one and up to 1.6 $m^3$ (56.4 $ft^3$) more in the third year after treatment. In all three post-monitoring years, mean wood volumes in the treatment reach were more than that of the reference reach (Figure 2).

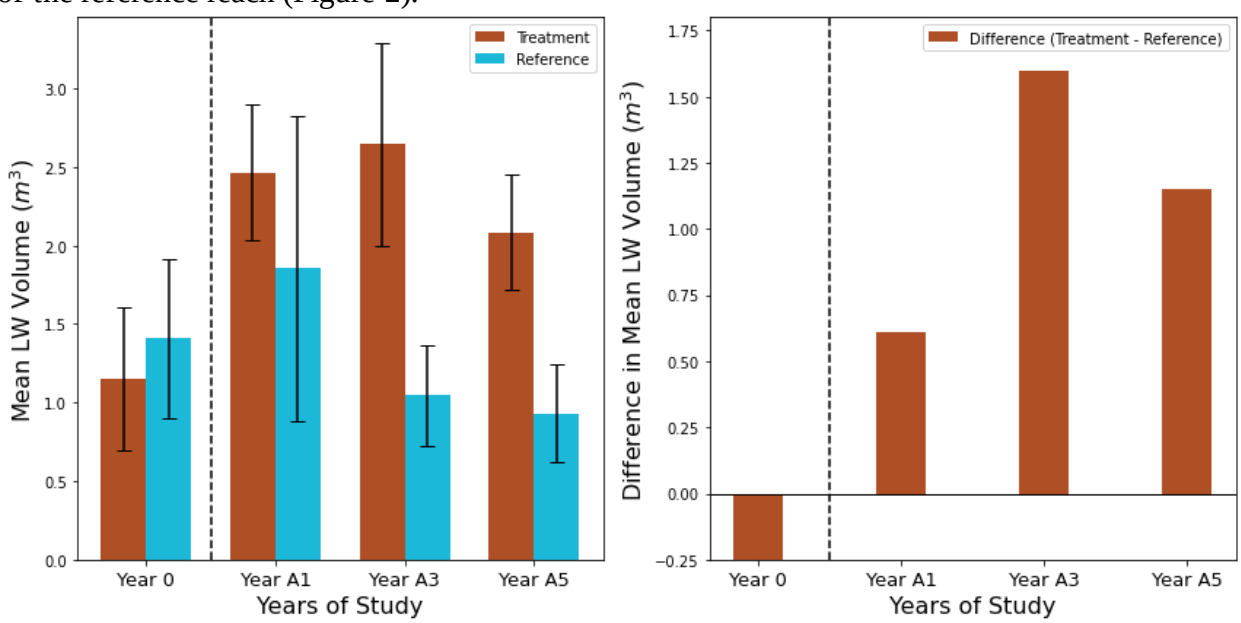


Figure 2. Mean large wood (LW) volume ($m^3$) measured within each reach over the study period, depicted for each year (left plot). Error bars represent one standard error on each side of the mean. The second plot illustrates the yearly differences in mean large wood volume ($m^3$) within each reach (right plot). The dotted black line denotes the separation between the years of pre-monitoring (Year 0) and post-monitoring years (Year A1, A3, and A5).

## Thalweg Depth

Although wood loading increased after restoration, mean thalweg depth did not increase proportionally in the treatment reach (Figure 3). The treatment reach had greater thalweg depths

before treatment and those differences persisted after restoration (Figure 3). Overall, there were no significant changes in thalweg depth associated with the treatment base on the linear model BACI term analysis interactions (p = 0.8, 0.9, 0.9 for Year A1, A3, and A5 respectively).

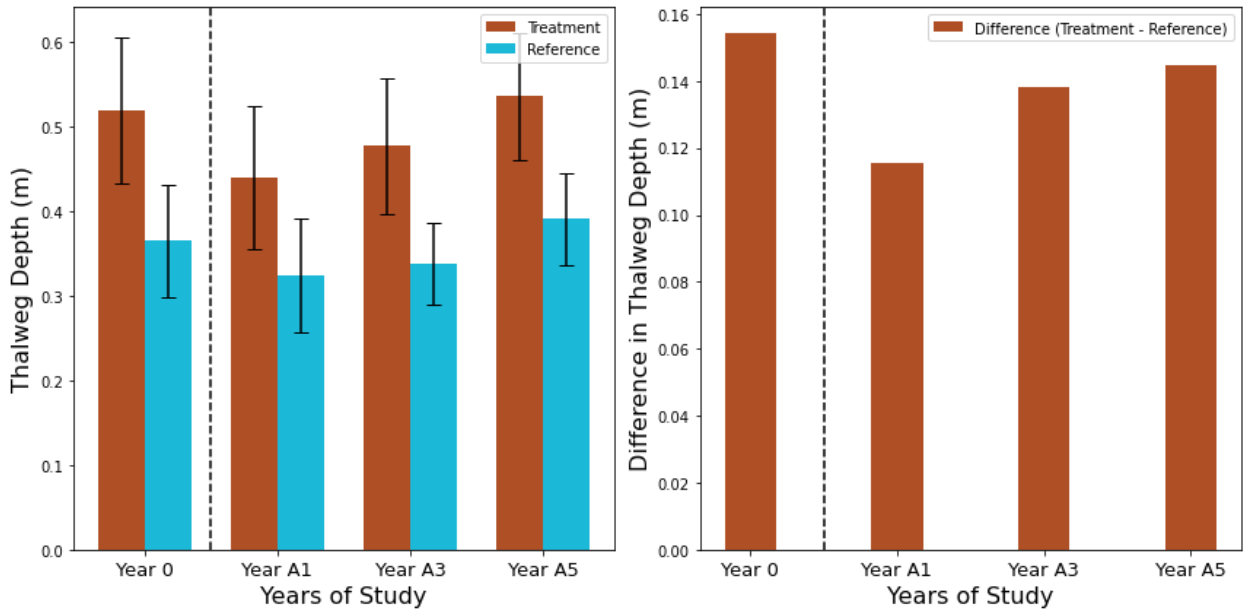


Figure 3. Mean thalweg depth (m) averaged across all transects within each reach over the study period, depicted for each year. Error bars represent one standard error on each side of the mean (left plot). The second plot illustrates the yearly differences in mean thalweg depth (m) within each reach (right plot). The dotted black line denotes the separation between the years of pre-monitoring (Year 0) and post-monitoring years (Year A1, A3, and A5).

## Pool area

Pool areas prior to restoration treatment were similar in both reaches. Large differences in pool area the first year after treatment (Year A1) were observed, with a greater change in pool area between these two years for the reference reach. The treatment reach consistently maintains a greater pool area than the reference reach during the three post-monitoring years (Year A1, A3, and A5). The difference in pool area between both reaches decreases as time passes, with the smallest difference observed in Year A5 (Figure 4).

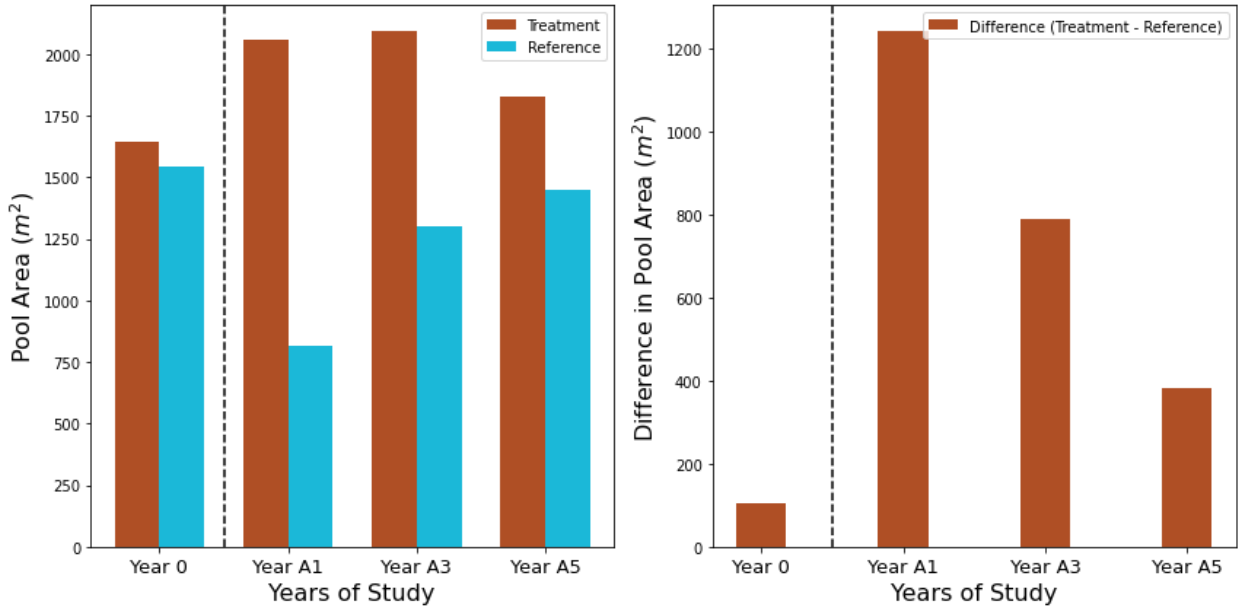


Figure 4. Total pool area ($m^2$) measured within each reach over the study period, depicted for each year (left plot). The second plot illustrates the yearly differences in total pool area ($m^2$) within each reach (right plot). The dotted black line denotes the separation between the years of pre-monitoring (Year 0) and post-monitoring (Year A1, A3, and A5).

## Juvenile coho salmon and all juvenile salmonids abundance

The estimated total abundance of YOY coho salmon in pools was greater in the treatment reach before restoration relative to the reference reach (Figure 5). This difference increased substantially following the first and third years after restoration. In the pre-treatment periods, the treatment reach had 43% more estimated YOY coho salmon in pools than the reference reach, but after restoration, this increased to over double and four times for Year A1 and A3 respectively (Figure 5). However, in the Year A5 July survey, there was no relative increase in YOY Coho salmon and an estimated 30% fewer YOY coho salmon in treatment reach pools relative to reference reach pools in that year (Figure 5).

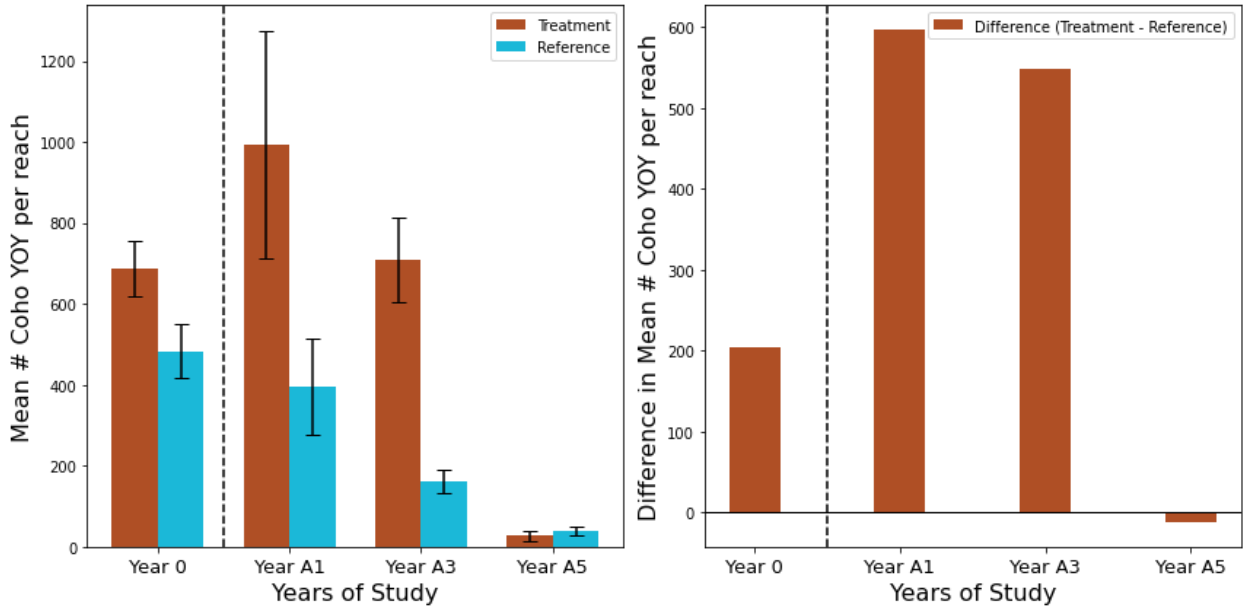


Figure 5. Mean estimates of the total number of Coho YOY within the treatment and reference reach across the five years of the study (left plot) compared to the difference of those estimates within each year of the study (right plot). The dotted black line denotes the separation between the years of pre-monitoring (Year 0) and post-monitoring (Year A1, A3, and A5). Error bars show one estimated standard error around each abundance estimate.

When considering all YOY salmonids results, their results varied among years. The treatment reach had greater total YOY salmonids in pools in all years, but the total and relative difference between reference and treatment reach varied. The year with the greatest total difference between reference and treatment reach YOY salmonids abundances was Year A1. However, the greatest proportional difference between the treatment and reference reach was in Year A3 at four times the estimated total salmonids while in Year A1, there was only triple the estimated total YOY salmonids. In Year A5, the total difference was the smallest, but the total estimated number of YOY salmonids was also smallest (Figure 6).

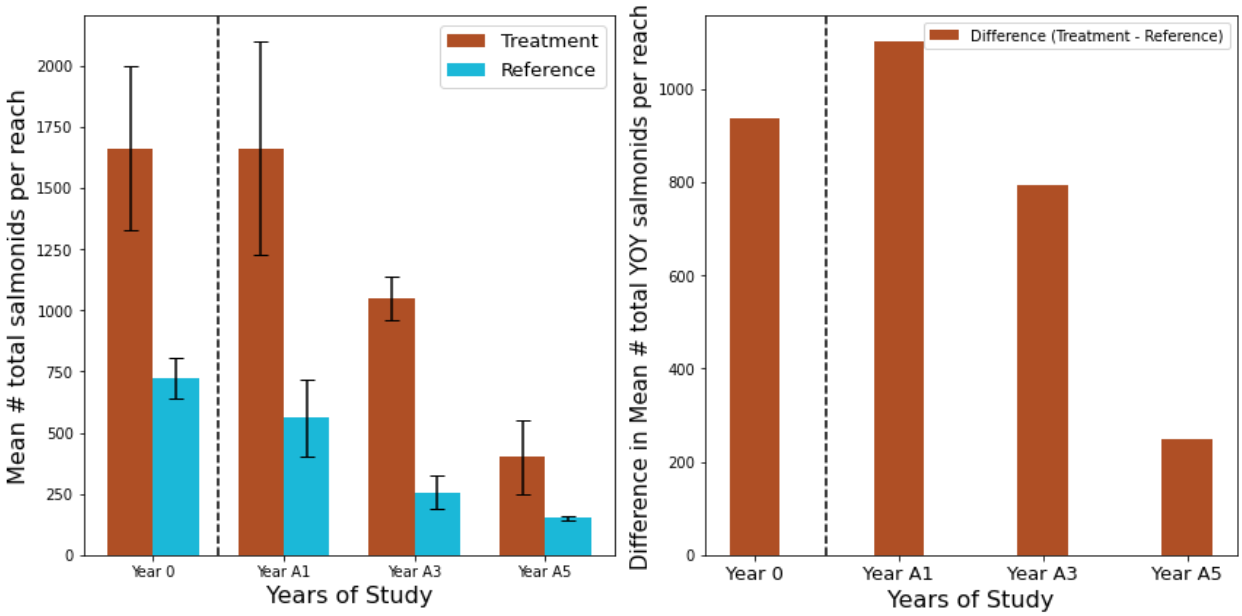


Figure 6. Mean estimates of the total number of young of year (YOY) salmonids within the treatment and reference reach across the five years of the study (left plot) compared to the difference of those estimates within each year of the study (right plot). The dotted black line denotes the separation between the years of pre-monitoring (Year 0) and post-monitoring (Year A1, A3, and A5). Error bars show the estimated standard error around each abundance estimate.

# Discussion

This study assessed the response of juvenile coho salmon and stream habitat to large wood restoration on a coastal stream in northern California using a BACI study design. We found that overall pool area increased following wood restoration, but that the increases declined in year A5 relative to years A1 and A3 post-restoration. Although the pool area increased, stream thalweg depths did not change in response to this restoration. Juvenile coho salmon had strong initial increases following restoration, however, they did not persist and in year A5 where the surveys indicated fewer juvenile coho in the treatment reach but abundances in both reaches were substantially lower than in other years. Overall, this study suggests that large wood addition in these systems does increase key habitat for threatened salmonid species, but long-term effects remain uncertain.

Other wood addition studies have found positive responses in pool area to wood addition. In this study, the total volume of wood doubled, and pool area increased by 25% in the first year following restoration treatment. Although a positive response was observed, it is important to note that not all wood configurations function to create pools (Crispin et al. 1993, Pess et al. 2023). The increase in pool area from our study aligns with other recent wood studies where increases in pool areas have resulted in increased density of coho salmon abundance (Roni et al. 2015). The pool area increases in year A1 persisted through year A3 and declined only slightly in year A5 in the treatment reach. When accounting for changes in pool area at the reference reach over time in the BACI analysis, the relative response in pool area showed a more dramatic increase in year A1 and year A3. In the fifth year after assessment (A5) there was still greater

pool area at the treatment reach – in total compared to pre-treatment conditions and the relative to the reference reach – but the relative increase was diminished compared to years A1 and A3 after the BACI analysis because pool area increased in the reference reach from year A3 to A5.

The lack of a response in mean thalweg depth was surprising since pool area increased, and wood is known to promote scour (Wohl and Scott 2017). However, the thalweg metric here was a mean of multiple transects within the reach and some of those may have accumulated what was scoured from other patches in the reach. The standard deviation of thalweg depth among transects decreased slightly from 0.274 m (0.900 ft) before treatment to 0.266 m (0.872 ft) after restoration in the treatment reach. And there was no change in the reference (0.212 m (0.700 ft) to 0.212 m (rounded)) suggesting that this shift in sediment could have been happening, but these changes are small and no change in overall depth is likely.

Total wood increased as expected in the complete census of wood. But the analysis which looked at reach-by-reach values had high wood variation among reaches, and this variation led to no significant increases in year A1 and A5 and only moderately significant increases in year A3. This limited significance in the analysis does not indicate a lack of added wood but rather highlights the spatial variation in how wood was arranged in each of these systems (i.e., wood was aggregated, and patchy and so variation was large).

Juvenile coho salmon response was complicated. Years A1 and A3 showed increases associated with increased pool area. This result is consistent with other studies targeting salmonid species and especially juvenile coho salmon who have been widely observed to have strong preference for pool habitats and particularly pools with large wood structures (Cederholm et al. 1997, Roni et al. 2015, Foote et al. 2020). However, in year A5, there was a shift with a drastic decrease in the juvenile coho abundance estimate, with slightly more estimated juvenile coho salmon in the reference reach relative to the treatment, prompting a need to understand the underlying reasons for this unexpected change. Mention values being so much smaller this year so the relative amount of different between reaches is small.

Weather events between year A3 and A5 could have caused large wood to migrate further downstream or even out of the treatment reach. Reducing the amount of wood or changing its location could decrease scour, which is an indicator of pool habitat and is important for juvenile coho salmon. Mean LW volume in the treatment reach, reference reach, and the relative difference both reaches decreased, less wood could mean less pool habitat. However, the BACI study design's strength is seen within comparing relative changes and should account for these weather-imposed differences. Another possibility is seen within the time of the sampling period across the post-monitoring years. In most years, snorkel surveys were conducted in the late summer (no earlier than mid-August and until mid-September). Year A5 survey, however, was conducted in July, three to six weeks before any other surveys. This inconsistency within sampling protocol could have resulted in an extremely low count during the survey and a relative difference between the treatment and reference reach. Additionally, since the surveying was only conducted during summer-time months, variation in winter rearing survival due to these large wood additions were not documented.

Furthermore, given that this study only encompasses one location with two reaches, this limited amount of replication can pose difficulties when determining significant relative change. Still, the data collected from all three post-monitoring years provides useful information regarding juvenile coho salmon responses to large wood additions.

## Conclusion

Long-term monitoring on scales larger than single large wood structures is crucial for salmonid habitat considering the time it takes for salmon to return to their natal streams. Additionally, the dynamic nature of streams and the fish they support underscores the importance of accurate reference reaches for assessing restoration efforts. By strategically implementing restoration efforts in critical areas and conducting sufficient post-treatment monitoring, practitioners can effectively evaluate changes in Pacific salmon abundances. Given inherent variation among streams, differences in restoration techniques and the many other controls on salmon, no single study alone can answer the question of whether wood addition broadly yields a benefit to threatened salmon. But individual studies are critical to this effort. To fully understand the complexities of large restoration efforts and their overall effectiveness, a wide range of well-constrained and rigorous long-term studies are needed. Overall, in the current study, we found no change in mean thalweg depth or depth variation, but we did find an increase in pool area. Based on snorkel surveys in a subset of pools during each year, greater pool area contributed to an increase in the density of juvenile coho salmon in the treatment reach in late summer in the first three years after wood addition. However, this did not hold through the fifth year perhaps due to overall low abundances in both reaches in year A5. This work therefore suggests that wood additions were meeting some aspects of core restoration goals, but that not all changes were met and not all responses persisted.

Collective results across many studies can help to develop a broader understanding of where and under what conditions a system may respond to as intended. However, for our efforts to have collective value, all results in a study – whether in line with or deviating from expected outcomes – must be reported. Given the magnitude of investments in restoration efforts, there is the potential for publication bias in which assessments that do not indicate the target response are not written-up or published (Foote et al. 2020). But reporting on non-results or contrasting results is critical to ensuring that restorations in the future can be done with the most comprehensive foundation of scientific data. In this study, demonstrating an unexpected lack of response in thalweg depth and the lack of response for juvenile coho salmon in year A5 is therefore just as important and demonstrating the expected increase in pool area and estimated juvenile coho salmon increases in year A1 and year A3 after restoration. This work makes important contributions about how a system responds to wood addition restoration, but this is one reach in one system. Scaling up to other places from this study alone should be done with caution and constrained to similar sized systems with comparable geomorphology.

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